Omid Arefian

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Employment History

Jun 2023 – Feb 2024 **Research Assistant, Shahid Beheshti University (Part-time), Tehran, Iran**
Research assistant to Professor Roghayeh Gavagsaz-Ghoachani in the Energy Engineering Laboratory.
• Leveraged experience to guide undergraduate students, enhancing their academic progression.
• Prepared six conference posters for international and domestic conferences.
• Collaborated with master's and doctoral students to foster knowledge sharing.
• Developed a comprehensive understanding of article publishing and research workflows.
• Gained expertise in photovoltaic panels technologies.

Jun 2022 – Sep 2022 **Summer Intern, Khorasan Steel Complex (Full-time), Mashhad, Iran**
Member of the repair line support team for the rebar rolling mill.
• Assisted senior technicians with routine equipment inspections.
• Helped perform basic maintenance and part replacements to keep the line running.
• Observed troubleshooting procedures and learned safe working practices in an industrial setting.
• Gained hands-on exposure to steel production workflows and machinery operation.

Education

2024 – Present **M.Sc. in Mechanical Engineering (Mechatronics & Robotics), Politecnico di Milano**
Cumulative GPA: 24.88/30

2019 – 2024 **B.Sc. in Mechanical Engineering, Shahid Beheshti University, Tehran**
GPA: 17.77/20

Projects

Fall 2025 **Advanced Nonlinear Control Design, Politecnico di Milano, Politecnico di Milano**
• Modeled a vehicle in a 2D environment and computed optimal obstacle-avoiding trajectories using direct and indirect methods.
• Design of Model Predictive Control (MPC) and Linear Quadratic Regulator (LQR) controllers to solve the optimal control problem.

Projects (continued)

Path-Following Controller design for Turtlebot3 in Gazebo using MATLAB and ROS2, Politecnico di Milano
• Integrated ROS2 with MATLAB under Ubuntu, enabling seamless simulation-to-hardware bridging.
• Designed and validated linear and nonlinear controllers (PID, MPC) for precise path tracking in cluttered environment.

Summer 2025 — **Custom Reinforcement-Learning Environment & Agent, Politecnico di Milano**
• Developed a PPO-based reinforcement learning simulation to coordinate multiple boxes, preventing collisions and arranging them in a straight line.
• Designed the agent to compute and apply optimal linear and angular velocity commands in a physics-driven environment.
• Demonstrated consistent collision avoidance and reliable alignment across varied initial conditions.

Spring 2025 — **System Identification of a Steel Plate, Politecnico di Milano**
• Conducted spectral FRF testing on a plate, modelling on Ansys to find the modal parameters.
• Implemented H1 and H2 estimators with FFT-based normalization and windowing.
• Developed a model-reduction workflow to approximate the plate's multi-degree-of-freedom (MDOF) dynamics by equivalent single-degree-of-freedom (SDOF) systems.
• Evaluated coherence and resonance features to select the most reliable FRF approach.

Experimental Modal Analysis of Light-Rail Wheel, Politecnico di Milano
• Configured a free–free wheel test using a dynamometric impact hammer and 12 piezoelectric accelerometers.
• Computed FRFs and coherence in MATLAB to extract the' natural frequencies, damping ratios and mode shapes of the axial modes.
• Employed MATLAB-based fitting methods to refine FRF models against experimental measurements.

Skills

Languages — English (C1), Italian (A2).
Software & Tools — MATLAB, Python, ANSYS, ROS 2, SolidWorks.

Research Interests

- Motion Planning
- Navigation

- Model Predictive Control
- Adaptive Control

Awards and Honors

- 2019–2023: Full Bachelor's Scholarship at Shahid Beheshti University, awarded for outstanding academic merit.
- 2019: Top 1 % in National University Entrance Exam, ranking among the very best participants.